

Putnam 2024 Solutions

Session A

- A1. Determine all positive integers n for which there exists positive integers a , b , and c satisfying

$$2a^n + 3b^n = 4c^n.$$

Answer. $n = 1$.

Solution. When $n = 1$, we may have $(a, b, c) = (1, 2, 2)$. To show that this is the only answer, denote $\nu_p(n)$ as the largest power of prime p dividing n . We note that if $\nu_p(a) \neq \nu_p(b)$ then $\nu_p(a + b) = \min\{\nu_p(a), \nu_p(b)\}$. Let $n \geq 2$, and $\nu_2(2a^n) = n\nu_2(a) + 1$; $\nu_2(3b^n) = n\nu_2(b)$, $\nu_2(4c^n) = 2 + n\nu_2(c)$, therefore $\nu_2(2a^n) \neq \nu_2(3b^n)$ at all times, and so $\min\{n\nu_2(a) + 1, n\nu_2(b)\} = 2 + n\nu_2(c)$. Considering congruence modulo n , the only possibility is $n = 2$.

When $n = 2$, w.l.o.g. assume that (a, b, c) not all divisible by 3, otherwise we can just change $(a, b, c) \rightarrow (\frac{a}{3}, \frac{b}{3}, \frac{c}{3})$. If $3 \nmid a$, then $2a^2 \equiv 2 \pmod{3}$, but $4c^2 \in \{0, 1\} \pmod{3}$. Therefore $3 \nmid b$ and $3 \mid a, c$. But this means $9 \nmid 2a^2 + 3b^2$ but $9 \mid 4c^2$, which is still a contradiction.

- A2. For which real polynomials p is there a real polynomial q such that

$$p(p(x)) - x = (p(x) - x)^2 q(x)$$

for all real x ?

Answer.

Solution. To show that each of these answers work, we note that if $p(x) = x + c$ for some c then $p(p(x)) = x + 2c$ so we may take $q(x) = \frac{2}{c}$ if $c \neq 0$, and 0 if $c = 0$. If $p(x) = -x + c$ then $p(p(x)) = p(-x + c) = -(-x + c) + c = x$, i.e. we may take $q(x) = 0$.

Now to show that these are the only answers, we first identify a number x_0 such that $p(x_0) = x_0$; such a number exists if $p(x) \neq x + c$ for some $c \neq 0$. Suppose that k is the highest power of $x - x_0$ dividing $p(x) - x$. Consider, now, a polynomial r such that $p(x_0 + \epsilon) - (x_0 + \epsilon) = \epsilon^k r(\epsilon)$ where $r(0) \neq 0$. For the problem statement to hold, we require $(x - x_0)^{2k}$ to divide $p(p(x)) - x$, or equivalently, $p(p(x_0 + \epsilon)) - (x_0 + \epsilon) \in O(\epsilon^{2k})$. Now,

$$p(p(x_0 + \epsilon)) = p(x_0 + (\epsilon + \epsilon^k r(\epsilon))) = x_0 + (\epsilon + \epsilon^k r(\epsilon) + (\epsilon + \epsilon^k r(\epsilon))^k r(\epsilon + \epsilon^k r(\epsilon)))$$

so $p(p(x_0 + \epsilon)) - (x_0 + \epsilon)$ is the following:

$$\epsilon^k [r(\epsilon) + (1 + \epsilon^{k-1} r(\epsilon))^k r(\epsilon + \epsilon^k r(\epsilon))]$$

If $k > 1$, as $\epsilon \rightarrow 0$ we have the product term as $2r(0) \neq 0$, so $p(p(x_0 + \epsilon)) - (x_0 + \epsilon)$ cannot be divisible by ϵ^{2k} ; (and the problem statement cannot hold). If $k = 1$, then the product term is $r(0) + (1 + r(0))r(0) = r(0)(2 + r(0))$. Given that $r(0) \neq 0$, we require $r(0) = -2$. Thus for all x_0 such that $p(x_0) - x_0 = 0$ we have $p(x) - x = (x - x_0)r(x)$ with $r(x_0) = -2$. This only holds when $p(x)$ is linear, i.e. in the form of $p(x) = -x - c$.

- A4. Find all primes $p > 5$ for which there exists an integer a and an integer r satisfying $1 \leq r \leq p-1$ with the following property: the sequence $1, a, a^2, \dots, a^{p-5}$ can be rearranged to form a sequence $b_0, b_1, b_2, \dots, b_{p-5}$ such that $b_n - b_{n-1} - r$ is divisible by p for $1 \leq n \leq p-5$.

Answer. $p = 7$.

Solution. When $p = 7$, we may take $a = 3$ and rearrange $1, 3, 3^2$ into $1, 9, 3$ and take $r = 1$.

To show $p = 7$ is the only solution, we first see that $b_0, \dots, b_{p-5}$ are pairwise distinct modulo p , and therefore same goes to the sequences $1, a, \dots, a^{p-5}$. Therefore numbers $a^k \not\equiv 1$ for $1 \leq k \leq p-5$. If $p \geq 11$, this means a is a primitive root of p (otherwise $a^k \equiv 1$ for some $k \leq \frac{p-1}{2} \leq p-5$). This means that $\{1, a, \dots, a^{p-2}\} = \{1, 2, \dots, p-1\}$ modulo p , i.e. if $a^{p-4}, a^{p-3}, a^{p-2} = x, y, z$ in that order then $\{b_0, \dots, b_{p-5}\} \sqcup \{x, y, z\} = \{1, 2, \dots, p-1\}$.

Now we may continue the sequence $b_{p-4}, b_{p-3}, b_{p-2}, b_{p-1}$, that together with $b_0, \dots, b_{p-5}$ form a complete residue modulo p . It follows that $\{b_{p-4}, b_{p-3}, b_{p-2}, b_{p-1}\} = \{x, y, z, 0\}$. Note from before that $y^2 = xz$, and depending on the order of x, y, z relative to 0, we have $\{x, y, z\}$ either equals to $\{s, 2s, 3s\}$ or $\{-s, s, 2s\}$ for some nonzero s . In the first case we either have $2^2 = 1 \cdot 3, 3^2 = 1 \cdot 2, 1^2 = 2 \cdot 3$, i.e. $p = 1, 7, 5$, respectively. In the second case we have $1^2 = 1 \cdot 2, 1^2 = -1 \cdot 2, 2^2 = -1 \cdot 1$, giving $p \leq 5$.

- A5. Consider the circle Ω with radius 9 and center at the origin $(0, 0)$, and a disk Δ with radius 1 and center at $(r, 0)$, where $0 \leq r \leq 8$. Two points P and Q are chosen independently and uniformly at random on Ω . Which value(s) of r minimize the probability that the chord $\overline{PQ}$ intersects Δ ?

Answer. $r = 0$.

Solution. Let Ω, Δ to have diameters D, d , respectively. (Hence $d = 2$ and $D = 18$ in our setting). We consider the following sampling of P and Q : sample a line $\vec{\lambda}$ through the origin O , set $PQ \parallel \vec{\lambda}$, and then sample $\theta = \angle POQ$ from OP to OQ (in $P \rightarrow Q$ fashion, hence modulo 2π). Then we're sampling λ uniformly, and conditioned on λ , we are also choosing $\theta \in [0, 2\pi)$ uniformly.

Next, fix λ and we calculate the probability p_λ that conditioned on $PQ \parallel \lambda$, what is the probability PQ intersecting Δ . To this end we set P_1Q_1, P_2Q_2 as the two chords satisfying the criterion $PQ \parallel \lambda$, and $P_1Q_1Q_2P_2$ in that order. Moreover P_1Q_1, P_2Q_2 are distinct and tangent to Δ . Then the $p_\lambda = \frac{\angle(P_1P_2)}{\pi}$ (i.e. proportional to the angle subtended by P_1P_2). Now that P_1Q_1 and P_2Q_2 always have distance d , we have $\angle(P_1P_2) = \arcsin \frac{d}{P_1Q_2} \geq \arcsin \frac{d}{D}$. Hence the desired probability (conditioned on p_λ , and overall) is always bounded below by $\frac{1}{\pi} \cdot \arcsin \frac{d}{D}$.

Equality holds if and only if P_1Q_2 is a diameter for all $\vec{\lambda}$. This happens if and only if the two circles are concentric; for other cases, conditioned on $\vec{\lambda}$, it holds if and only if $\lambda \parallel$ the two centers of the circles. This gives $r = 0$ as the only answer.

Session B

- B1. Let n and k be positive integers. The square in the i th row and j th column of an n -by- n grid contains the number $i + j - k$. For which n and k is it possible to select n squares from the grid, no two in the same row or column, such that the numbers contained in the selected squares are exactly $1, 2, \dots, n$?

Answer. All n odd and $k = \frac{n+1}{2}$.

Solution. Suppose that the condition can be fulfilled, and suppose that for each $i = 1, \dots, n$ we choose $c(i)$ -th element from row i . Then

$$\frac{n(n+1)}{2} = \sum_{i=1}^n i = \sum_{i=1}^n (i + c(i) - k) = \frac{n(n+1)}{2} + \frac{n(n+1)}{2} - kn.$$

Thus $kn = \frac{n(n+1)}{2}$ and $k = \frac{n+1}{2}$ is necessarily, so n is odd.

For construction, consider the following entries:

$$(1, k), (2, k+1), \dots, (k, 2k-1), (k+1, 1), \dots, (2k-1, k-1)$$

noting that $2k-1 = n$. Then the sum above is now $1, 3, \dots, n, 2, 4, \dots, n-1$, as desired.

- B2. Two convex quadrilaterals are called partners if they have three vertices in common and they can be labeled $ABCD$ and $ABCE$ so that E is the reflection of D across the perpendicular bisector of the diagonal $\overline{AC}$. Is there an infinite sequence of convex quadrilaterals such that each quadrilateral is a partner of its successor and no two elements of the sequence are congruent?

Answer. No.

Solution. We note that after each operation, the area of the convex quadrilateral remains the same, and the 4 side lengths also remain the same (except in possibly different orders). Given that there are at most $4! = 24$ ordering of the sides, it suffices to show that there are only finitely many quadrilaterals of given side lengths and area.

(TODO)

- B3. Let r_n be the n th smallest positive solution to $\tan x = x$, where the argument of tangent is in radians. Prove that

$$0 < r_{n+1} - r_n - \pi < \frac{1}{(n^2 + n)\pi}$$

for $n \geq 1$.

Solution. We note that $\tan x$ is positive only when there exists an integer k with $x \in (k\pi, (k + \frac{1}{2})\pi)$. In addition, differentiating $\tan x - x$ gives us $\tan^2 x \geq 0$ so there is exactly one intersection of $\tan x$ and x for each interval $(k\pi, (k + \frac{1}{2})\pi)$. Thus in other words $r_n \in (n\pi, (n + \frac{1}{2})\pi)$. Now note also that $\tan r_{n+1} > \tan r_n$. Let $s_n = r_n - \pi$, we have $\tan s_{n+1} = r_{n+1} > r_n = \tan s_n$ with $s_n, s_{n+1} \in (0, \frac{\pi}{2})$. Given that $\tan$ is increasing in this interval, $s_{n+1} > s_n$ and so $r_{n+1} - r_n > \pi$.

To prove the other side of the inequality, we now show that $\tan s_n = \tan r_n > \pi\sqrt{n^2 + n}$. Now $\tan s_n = s_n + n\pi$; on the other hand we have $\tan s_n = \frac{1}{\tan(\frac{\pi}{2} - s_n)} < \frac{1}{\frac{\pi}{2} - s_n}$, using the fact that $\tan x > x$ for $x \in (0, \frac{\pi}{2})$. Therefore we are now solving $x + n\pi < \frac{1}{\frac{\pi}{2} - x}$, or $(x + n\pi)(\frac{\pi}{2} - x) < 1$. Given also that $(x + n\pi)(\frac{\pi}{2} - x)$ is decreasing in $(0, \frac{\pi}{2})$, it suffices to show that $(x + n\pi)(\frac{\pi}{2} - x) > 1$ if $x = \pi\sqrt{n(n+1)} - n\pi$. Indeed, for this x we have

$$\frac{\pi}{2} - x = \pi(n + \frac{1}{2} - \sqrt{n(n+1)}) = \pi(\sqrt{n^2 + n + \frac{1}{4}} - \sqrt{n^2 + n}) = \frac{\pi}{4(\sqrt{n^2 + n + \frac{1}{4}} + \sqrt{n^2 + n})}$$

where the last equality follows from $\sqrt{a+b} - \sqrt{a} = \frac{b}{\sqrt{a+b} + \sqrt{a}}$. Thus putting them together we have

$$(x + n\pi)(\frac{\pi}{2} - x) = \pi^2 \frac{\sqrt{n^2 + n}}{4(\sqrt{n^2 + n + \frac{1}{4}} + \sqrt{n^2 + n})} = \pi^2 \frac{1}{4(1 + \sqrt{\frac{n^2 + n + \frac{1}{4}}{n^2 + n}})} \geq \pi^2 \frac{1}{4(1 + \sqrt{\frac{9}{8}})} > 1$$

where the last part following $n \geq 1$ and using $\pi^2 > 9$ and $\sqrt{\frac{9}{8}} < \frac{5}{4}$. Thus the claim is proven. With this, all it remains is the fact that

$$\pi = (\tan s_{n+1} - s_{n+1}) - (\tan s_n - s_n) = \int_{s_n}^{s_{n+1}} \tan^2 x dx$$

and since $\tan x > \pi\sqrt{n^2 + n}$ for all $x \in (s_n, s_{n+1})$ we have $s_{n+1} - s_n < \frac{\pi}{\pi^2(n^2 + n)}$, as desired.

- B4. Let n be a positive integer. Set $a_{n,0} = 1$. For $k \geq 0$, choose an integer $m_{n,k}$ uniformly at random from the set $\{1, \dots, n\}$, and let

$$a_{n,k+1} = \begin{cases} a_{n,k} + 1, & \text{if } m_{n,k} > a_{n,k}; \\ a_{n,k}, & \text{if } m_{n,k} = a_{n,k}; \\ a_{n,k} - 1, & \text{if } m_{n,k} < a_{n,k}. \end{cases}$$

Let $E(n)$ be the expected value of $a_{n,n}$. Determine $\lim_{n \rightarrow \infty} E(n)/n$.

Answer. $\frac{1}{2}(1 - e^{-2})$.

Solution. Let $b_{n,k} = a_{n,k} - \frac{n+1}{2}$; we show that $\mathbb{E}[b_{n,k+1}] = (1 - \frac{2}{n})\mathbb{E}[b_{n,k}]$. Indeed, the probability of $b_{n,k+1} = (b_{n,k} - 1, b_{n,k}, b_{n,k} + 1)$ is $(\frac{a_{n,k}-1}{n}, \frac{1}{n}, 1 - \frac{a_{n,k}}{n})$, i.e. $(\frac{b_{n,k}-1/2}{n} + \frac{1}{2}, \frac{1}{n}, \frac{1}{2} - \frac{b_{n,k}+1/2}{n})$, giving

$$\mathbb{E}[b_{n,k+1}|b_{n,k} = b] = (b-1)(\frac{1}{2} + \frac{b-1/2}{n}) + \frac{b}{n} + (b+1)(\frac{1}{2} - \frac{b+1/2}{n}) = b(1 - \frac{2}{n})$$

Thus the claim follows from linearity of expectation. Given that $b_{n,0} = 1 - \frac{n+1}{2} = -\frac{n-1}{2}$, we have

$$\mathbb{E}[b_{n,n}] = -\frac{n-1}{2}(1 - \frac{2}{n})^n$$

and therefore

$$\frac{E(n)}{n} = \frac{n+1}{2n} + \frac{\mathbb{E}[b_{n,n}]}{n} = \frac{n+1}{2n} - \frac{n-1}{2n}(1 - \frac{2}{n})^n \rightarrow \frac{1}{2} - \frac{1}{2}(e^{-2})$$

- B5. Let k and m be positive integers. For a positive integer n , let $f(n)$ be the number of integer sequences $x_1, \dots, x_k, y_1, \dots, y_m, z$ satisfying $1 \leq x_1 \leq \dots \leq x_k \leq z \leq n$ and $1 \leq y_1 \leq \dots \leq y_m \leq z \leq n$. Show that $f(n)$ can be expressed as a polynomial in n with nonnegative coefficients.
- B6. For a real number a , let $F_a(x) = \sum_{n \geq 1} n^a e^{2n} x^{n^2}$ for $0 \leq x < 1$. Find a real number c such that

$$\begin{aligned} \lim_{x \rightarrow 1^-} F_a(x) e^{-1/(1-x)} &= 0 \quad \text{for all } a < c, \text{ and} \\ \lim_{x \rightarrow 1^-} F_a(x) e^{-1/(1-x)} &= \infty \quad \text{for all } a > c. \end{aligned}$$

Answer. $c = -\frac{1}{2}$.

Solution. Let $M = \frac{1}{\log \frac{1}{x}}$; note that $x \rightarrow 1^-$ means $M \rightarrow +\infty$. Then for each n we have

$$e^{2n} x^{n^2} = e^{2n - \frac{n^2}{M}} = e^{M - \frac{(n-M)^2}{M}}$$

In addition, we have

$$e^{1/(1-x)} = e^{1/(1-e^{-1/M})} = e^{M/(1-O(1/M))} = e^{M+O(1)}$$

as $M \rightarrow \infty$. Thus it suffices to consider the quantity

$$\sum_{n \geq 1} n^a e^{-\frac{(n-M)^2}{M}}$$

Thus it follows that

$$F_a(x) > \sum_{n \in (M-\sqrt{M}, M+\sqrt{M})} n^a e^{M - \frac{(n-M)^2}{M}} > e^{M-1} 2 \lfloor \sqrt{M} \rfloor \min\{(M - \sqrt{M})^a, (M + \sqrt{M})^a\}$$

For M sufficiently large, we have $\min\{(M - \sqrt{M})^a, (M + \sqrt{M})^a\} > \frac{1}{2}M^a$, giving the inequality as $> \frac{1}{2}e^{M-1}\sqrt{M} \cdot M^a = e^{M-1}M^{a+\frac{1}{2}}$. Note also that $e^{-1/(1-x)} = e^{-1/(1-e^{-\frac{1}{M}})} = e^{-(M+O(1))}$, so $F_a(x)e^{-1/(1-x)} > e^{-1+O(1)}M^{a+\frac{1}{2}}$, which is $\rightarrow \infty$ as $M \rightarrow \infty$, if $a > -\frac{1}{2}$.

Conversely, suppose now that $a < -\frac{1}{2}$. Denote $S_k = \sum_{n: (k-1)\sqrt{M} \leq |n-M| \leq k\sqrt{M}} n^a e^{2n} x^{n^2}$, we have $F_a(x) \leq \sum_{k \geq 1} S_k$. Notice that